FLAT AND ETALE DESCENT ALONG THE PUSHOUT SQUARE: THE AFFINE FLIP EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 19955-19980, immediately following the statement at 19945-19953.

Editorial note. The sentence recording that the comparison arrow need not be an isomorphism without flatness (source lines 19965-19971) is part of the counted proof and is the reason the flat hypothesis is essential.

Editorial note. Two cross-chapter pointer families (More on Algebra) are recorded in assembly.external_references; the Situation and the defining Lemma are same-chapter line-indexed stubs.

Allowed standard prerequisites (editorial).

- The pushout situation and its functor. Situation situation-pushout-along-closed-immersion-and-integral and Lemma lemma-pushout-functor of this chapter, cited at statement lines 19945-19948. Fixes the pushout diagram Y amalg_Z X (Z a closed subscheme of X with X -> Y integral) and the data tuple (X', Y', Z', i', j', f, g, h) whose category is being compared.
- Flatness and etaleness of the induced pushout morphism. More on Algebra, Lemma more-algebra-lemma-properties-algebras-over-fibre-product, cited at proof lines 19952-19954 and 19975-19977. Both directions hinge on it: forward, Y' amalg_{Z'} X' -> Y amalg_Z X is flat/etale; converse, the comparison arrow is an isomorphism exactly when W is flat.
- Closed immersion behaviour without flatness. More on Algebra, Lemma more-algebra-lemma-module-over-fibre-product-bis, cited at proof lines 19968-19971. Records that without flatness the comparison arrow is only a closed immersion in general, which is why the flat hypothesis cannot be dropped.

Source reference. Original lemma 0ECK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19876–19876. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08IG: focused-seeds/geometry/S9-morph-pushout-functor-equivalence-flat/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08KQ: focused-seeds/geometry/S9-morph-pushout-functor-equivalence-flat/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0ECI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19708–19708. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0ECI the category of schemes flat, separated, and locally quasi-finite over the pushout $Y \amalg_Z X$ is equivalent to the category of $(X', Y', Z', i', j', f, g, h)$ as in Lemma 0ECK with f, g, h flat. Similarly with "flat" replaced with "étale".*

Proof. If we start with $(X', Y', Z', i', j', f, g, h)$ as in Lemma 0ECK with f, g, h flat or étale, then $Y' \amalg_{Z'} X' \to Y \amalg_Z X$ is flat or étale by More on Algebra, Lemma 08KQ.

For the converse, let $W \to Y \amalg_Z X$ be a separated and locally quasi-finite morphism. Set $X' = W \times_{Y \amalg_Z X} X$, $Y' = W \times_{Y \amalg_Z X} Y$, and $Z' = W \times_{Y \amalg_Z X} Z$ with obvious morphisms i', j', f, g, h. Form the pushout $Y' \amalg_{Z'} X'$. We obtain a morphism

$$Y' \amalg_{Z'} X' \longrightarrow W$$

of schemes over $Y \amalg_X Z$ by the universal property of the pushout. If we do not assume that $W \to Y \amalg_Z X$ is flat, then in general this morphism won't be an isomorphism. (In fact, More on Algebra, Lemma 08IG shows the displayed arrow is a closed immersion but not an isomorphism in general.) However, if $W \to Y \times_Z X$ is flat, then it is an isomorphism by More on Algebra, Lemma 08KQ. □